

# SIEM LAB – SPLUNK ENTERPRISE

## SUMMARY

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Security Information and Event Management (SIEM) platforms play a critical role in modern Security Operations Centers (SOCs) by collecting, correlating, and analyzing security events from multiple systems. This project demonstrates the deployment of Splunk Enterprise as a centralized SIEM solution for monitoring Windows endpoints using native Windows Event Logs and Sysmon telemetry.

A Windows 10 endpoint was configured with the Splunk Universal Forwarder and Sysmon to collect Security, System, Application, and advanced endpoint telemetry. A realistic attack was simulated from a Kali Linux workstation involving network reconnaissance, SSH brute-force authentication, successful remote access, PowerShell execution, privilege escalation, and persistence through Scheduled Tasks and Registry Run Keys.

Custom detection rules, dashboards, and email alerts were developed to identify malicious activities. The investigation reconstructed the complete attack timeline and mapped all attacker techniques to the MITRE ATT&CK framework.

The project demonstrates practical experience in SIEM deployment, detection engineering, threat hunting, incident investigation, and SOC operations.

## OVERVIEW

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Organizations generate thousands of security events every day. Without centralized monitoring, malicious activities such as unauthorized logins, privilege escalation, and persistence techniques may remain undetected.

The objective of this project was to implement a centralized SIEM capable of detecting common attacker techniques and assisting analysts during incident investigations.

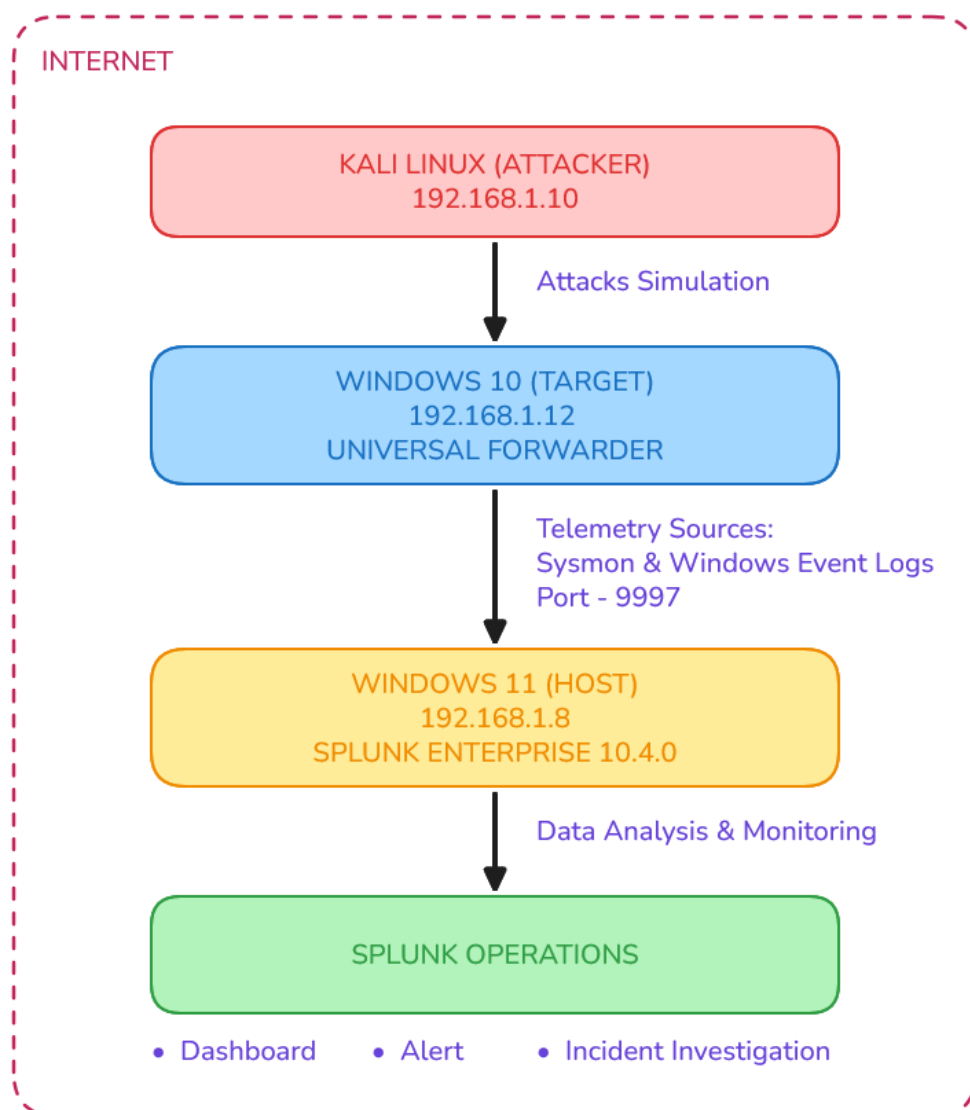
## OBJECTIVES

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- Deploy Splunk Enterprise as a SIEM.
- Configure Windows log forwarding.
- Deploy Sysmon for enhanced endpoint telemetry.
- Develop custom detection rules.
- Build a SOC monitoring dashboard.
- Configure automated email alerts.
- Simulate realistic attacks.
- Investigate the attack using Splunk.
- Map detections to MITRE ATT&CK.

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## LAB ARCHITECTURE



The lab consisted of three systems connected on the same local network.

| System     | Role        | IP Address   |
|------------|-------------|--------------|
| Kali Linux | Attacker    | 192.168.1.10 |
| Windows 10 | Victim      | 192.168.1.12 |
| Windows 11 | Splunk SIEM | 192.168.1.8  |

The Windows endpoint forwarded Windows Event Logs and Sysmon logs to the Splunk Enterprise server using the Splunk Universal Forwarder over TCP port 9997.

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## ENVIRONMENT SETUP

### 1. Splunk Enterprises Setup

#### 1.1 Download and Install Splunk Enterprises

Visit official Splunk Enterprises website and register for free trial. Download the latest [Splunk Enterprise](#) release (10.4.0 for me) on Windows 11 host.

Install the downloaded .msi file with default configuration.

- Accept Licence
- Local System Account
- Create Admin username/password
- Install

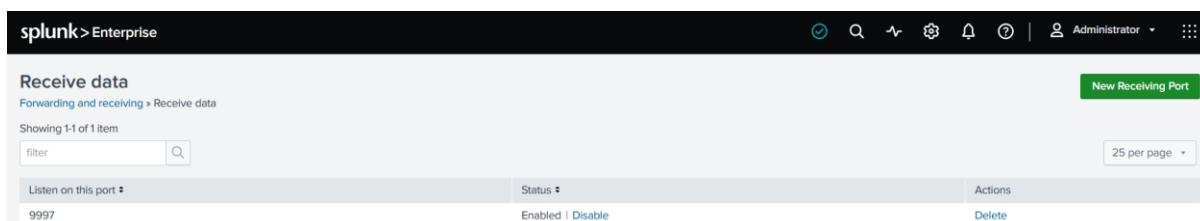
#### 1.2 Access and Setup Splunk Dashboard

Open any browser and visit <http://localhost:8000/> login with credential to access Splunk Dashboard.



Setup Receiving port on Splunk Instance.

Settings → Forwarding and Receiving → Receive Data → Configure Receiving



#### 1.3 Allow Splunk receiving port

Create a Windows defender firewall rule, to allow Inbound traffic over port 9997.

| Inbound Rules           |       |                 |         |        |          |               |               |                |          |            |
|-------------------------|-------|-----------------|---------|--------|----------|---------------|---------------|----------------|----------|------------|
| Name                    | Group | Profile         | Enabled | Action | Override | Program       | Local Address | Remote Address | Protocol | Local Port |
| ✓ Riot Client           |       | Private         | Yes     | Allow  | No       | C:\riot ga... | Any           | Any            | UDP      | Any        |
| ✓ Splunk Forwarder Port |       | Private, Public | Yes     | Allow  | No       | Any           | Any           | Local subnet   | TCP      | 9997       |
| ✓ Steam                 |       | All             | Yes     | Allow  | No       | C:\Progra...  | Any           | Any            | TCP      | Any        |

#### 1.4 Validate the open port on Windows 11

Open CMD and check for Port 9997 is open and ready to capture connection.



#### 1.5 Create Sysmon Index with default settings

| Name ↓ | Status ↓ | Type ↓   | App ↓  | Current Size ↓ | Max Size ⬆️ ↓ | Event Count ↓ | Earliest Event ↓ | Latest Event ↓    | Home Path ↓           |
|--------|----------|----------|--------|----------------|---------------|---------------|------------------|-------------------|-----------------------|
| sysmon | ● Active | 📁 Events | search | 7 MB           | 500 GB        | 23.2K         | 8 days ago       | a few seconds ago | \$SPLUNK_DB\sysmon\db |

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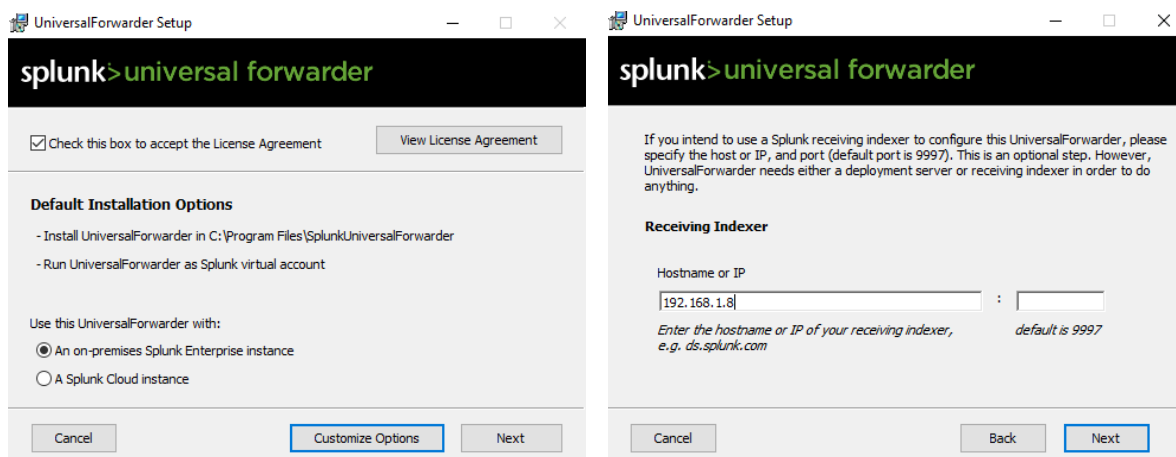
## 2. Spunk Universal Forwarder Setup

### 2.1 Download and Install Universal Forwarder

On windows 10 VM open any browser and navigate to Splunk Official website login with previously registered account and Download the latest version of [Universal Forwarder](#).

Install the downloaded .msi file on Windows 10 VM.

- Accept Licence Agreement
- Select On-Premise Splunk Enterprises Instance
- Setup Admin username and Password
- Deployment Server - Leave it blank.
- Receiving Indexer – Enter Windows 11 Host IP and Default port 9997



### 2.2 Download Sysmon and setup

Download the [SYSMON](#) from Microsoft official website and extract the contents from zip file to C:\Sysmon directory.

Download [SwiftOnSecurity Sysmon Configuration](#) file from Github, and extract all the contents in same C:\Sysmon directory.

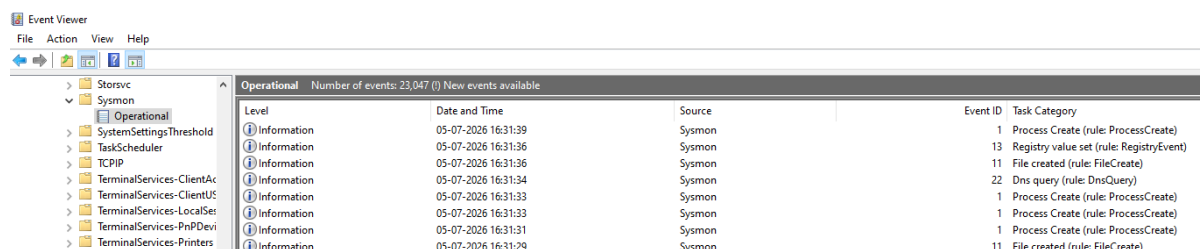
Open CMD as Administrator and Install Sysmon via commandline.

```
C:\Sysmon>sysmon64 -accepteula -i
```

Load the Downloaded Configuration File

```
C:\Sysmon>sysmon64 -c sysmonconfig-export.xml
```

Verify the Sysmon in Event Viewer



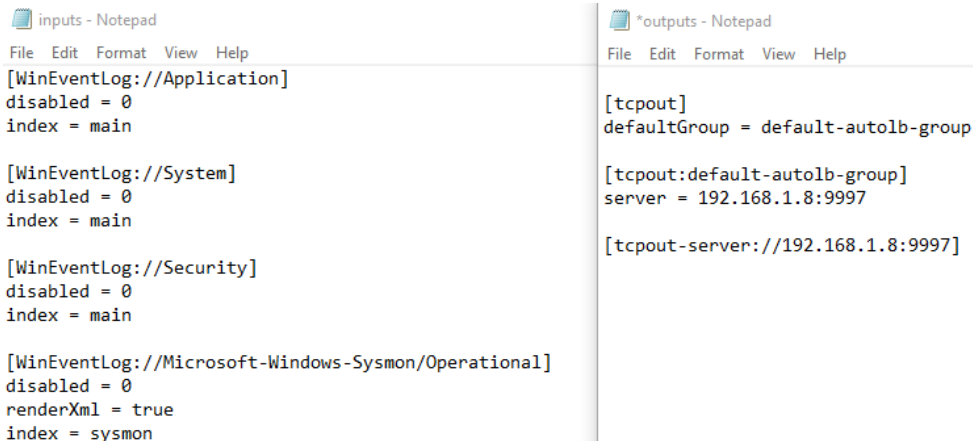
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## 3. Configure Universal Forwarder to send Windows and Sysmon Logs

Navigate to, C:\Program Files\SplunkUniversalForwarder\etc\system\local

If not present create inputs.conf file with following settings to enable logs flow.

Check the outputs.conf file for following settings.



```
inputs - Notepad
File Edit Format View Help
[WinEventLog://Application]
disabled = 0
index = main

[WinEventLog://System]
disabled = 0
index = main

[WinEventLog://Security]
disabled = 0
index = main

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
renderXml = true
index = sysmon

*outputs - Notepad
File Edit Format View Help

[tcpout]
defaultGroup = default-autolb-group

[tcpout:default-autolb-group]
server = 192.168.1.8:9997

[tcpout-server://192.168.1.8:9997]
```

Open CMD as Administrator and restart splunk forwarder.

```
C:\Program Files\SplunkUniversalForwarder\bin>splunk restart
```

Check for Forwarder Servers configured in outputs.conf

```
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Your session is invalid. Please login.
Splunk username: admin
Password:
Active forwards:
    192.168.1.8:9997
Configured but inactive forwards:
    None
```

Open PowerShell and check Forwarder Status

```
PS C:\Windows\system32> Get-Service -Name SplunkForwarder

Status      Name              DisplayName
-----
Running     SplunkForwarder   SplunkForwarder
```

Check Connection to Splunk Instance

```
PS C:\Windows\system32> Test-NetConnection 192.168.1.8 -port 9997

ComputerName      : 192.168.1.8
RemoteAddress     : 192.168.1.8
RemotePort        : 9997
InterfaceAlias    : Ethernet0
SourceAddress     : 192.168.1.12
TcpTestSucceeded  : True
```

On Windows check the established connection

```
C:\Users\Aditya>netstat -ano | findstr "9997"

TCP    0.0.0.0:9997        0.0.0.0:0         LISTENING        24620
TCP    192.168.1.8:9997  192.168.1.12:50845 ESTABLISHED      24620
```

Now Splunk setup is ready to capture logs.

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## LOG COLLECTION

Two indexes were configured within Splunk.

### Main Index

- Security Logs
- System Logs
- Application Logs

**New Search**  
index=main  
✓ 11,953 events (before 7/6/26 12:20:52.000 PM) No Event Sampling

**Index Name & Total Events**

**Available Log Sources**

| Values                  | Count | %      |
|-------------------------|-------|--------|
| WinEventLog:Security    | 9,034 | 75.57% |
| WinEventLog:System      | 1,780 | 14.89% |
| WinEventLog:Application | 1,139 | 9.52%  |

### Sysmon Index

- Process Creation
- Network Connections
- Registry Modifications

**New Search**  
index=sysmon  
✓ 24,612 events (before 7/6/26 12:24:30.000 PM) No Event Sampling

**Index Name and Total Events**

**Top 10 Values**

| EventCode | Count  | %      |
|-----------|--------|--------|
| 1         | 11,826 | 59.15% |
| 13        | 2,889  | 14.45% |
| 5         | 2,421  | 12.11% |
| 11        | 2,233  | 11.16% |
| 22        | 217    | 1.08%  |
| 3         | 208    | 1.04%  |
| 12        | 156    | 0.78%  |
| 255       | 16     | 0.08%  |
| 4         | 9      | 0.04%  |
| 2         | 7      | 0.03%  |

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## DETECTION

Detection focuses on creating, implementing, and validating detection rules that identify malicious activities from Windows Event Logs and Sysmon telemetry. Custom SPL queries were developed in Splunk Enterprise to detect common attacker techniques observed during the simulated attack. Each Detection was validated by performing events related activities.

| DETECTION                              | SOURCE                       | SPLUNK PROCESSING LANGUAGE QUERY                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Failed Logon Detection (4625)          | Windows Security Logs - 4625 | <pre>index=main LogName=Security EventCode=4625   timechart span=5m count</pre>                                                                                                                                                                                                       |
| Top Targeted User Accounts             | Windows Security Logs - 4625 | <pre>index=main LogName=Security EventCode=4625   stats count by user   sort -count   head 10</pre>                                                                                                                                                                                   |
| Successful Windows Logon (4624)        | Windows Security Logs - 4624 | <pre>index=main LogName=Security EventCode=4624 Logon_Type=2   search NOT user IN ("SYSTEM","LOCAL SERVICE","NETWORK SERVICE","defaultuser0")   where NOT like(user,"DWM-%")   where NOT like(user,"UMFD-%")   where NOT like(user,"%\$\$")   timechart span=5m count</pre>           |
| Successful Interactive Logon by users  | Windows Security Logs - 4624 | <pre>index=main LogName=Security EventCode=4624 Logon_Type=2   search NOT user IN ("SYSTEM","LOCAL SERVICE","NETWORK SERVICE","defaultuser0")   where NOT like(user,"DWM-%")   where NOT like(user,"UMFD-%")   where NOT like(user,"%\$\$")   stats count by user   sort -count</pre> |
| Recently Created User Account Count    | Windows Security Logs - 4720 | <pre>index=main LogName=Security EventCode=4720   stats count</pre>                                                                                                                                                                                                                   |
| Recently Created Local User Accounts   | Windows Security Logs - 4720 | <pre>index=main LogName=Security EventCode=4720   rename user as New_User   table _time host New_User   sort -_time</pre>                                                                                                                                                             |
| Administrator Group Membership Changes | Windows Security Logs - 4732 | <pre>index=main LogName=Security EventCode=4732 Group_Name=Administrators   rename Account_Name as Created_By   table _time host Created_By Group_Name   sort -_time</pre>                                                                                                            |

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| DETECTION                                         | SOURCE                 | SPLUNK PROCESSING LANGUAGE QUERY                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PowerShell Execution by User                      | Sysmon Event Code = 1  | <pre>index=sysmon EventCode=1 Image="*powershell.exe" AND CommandLine="*powershell*" NOT ParentImage="*splunkd.exe" NOT User="NT AUTHORITY\SYSTEM"   stats count by User   sort -count</pre>                                                                                                                                                      |
| Encoded PowerShell Commands Executions            | Sysmon Event Code = 1  | <pre>index=sysmon EventCode=1 Image="*powershell.exe" CommandLine="*-EncodedCommand*" NOT ParentImage="*splunkd.exe" NOT User="NT AUTHORITY\SYSTEM"   table _time User CommandLine   sort -_time</pre>                                                                                                                                            |
| PowerShell Spawned by Suspicious Parent Processes | Sysmon Event Code = 1  | <pre>index=sysmon EventCode=1 Image="*powershell.exe" ( ParentImage="*cmd.exe" OR ParentImage="*winword.exe" OR ParentImage="*excel.exe" OR ParentImage="*wscript.exe" OR ParentImage="*cscript.exe" OR ParentImage="*mshta.exe" OR ParentImage="*rundll32.exe" ) NOT ParentImage="*splunkd.exe"   table _time User ParentImage CommandLine</pre> |
| Recent Network Connections                        | Sysmon Event Code = 3  | <pre>index=sysmon EventCode=3   table _time SourceIp DestinationIp DestinationPort Protocol   sort -_time</pre>                                                                                                                                                                                                                                   |
| Network Connections by Process                    | Sysmon Event Code = 3  | <pre>index=sysmon EventCode=3 Image != "*OneDrive*"   stats count by Image   sort -count</pre>                                                                                                                                                                                                                                                    |
| Scheduled Task Creation Events                    | Sysmon Event Code = 1  | <pre>index=sysmon EventCode=1 Image="*schtasks.exe" CommandLine="*/create*"   table _time User ParentImage CommandLine   sort -_time</pre>                                                                                                                                                                                                        |
| Registry Run Key Persistence                      | Sysmon Event Code = 13 | <pre>index=sysmon EventCode=13 TargetObject="*CurrentVersion\\Run*"   table _time User Image TargetObject Details   sort -_time</pre>                                                                                                                                                                                                             |



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## DASHBOARD SETUP

A centralized dashboard was developed to provide continuous monitoring of the Windows endpoint using detection rules created earlier.

The dashboard contained different panels categorized into:

- Authentication
- Account Management
- PowerShell Monitoring
- Network Monitoring
- Persistence Monitoring

The dashboard enabled rapid identification of suspicious activities without manually searching individual logs.



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## ALERT CONFIGURATION

SMTP server was setup to receive the Alerts via Email.

Settings → Server Settings → Email Settings

The screenshot shows the 'Mail Server Settings' configuration page. It includes fields for 'Mail host' (smtp.gmail.com:587), 'Email security' (Enable TLS is selected), 'Authentication' (Simple is selected), 'Username' (adiitya.soc@gmail.com), and 'Email Domains' (Use custom domains is selected, with a list containing gmail.com).

Note – Enable 2 MFA and generate App Password from Gmail for SMTP setup

## Alert Creation

Settings → Searches, Reports and Alerts → New Alert

The screenshot shows the 'Create Alert' dialog box. It is divided into two main sections: 'Settings' on the left and 'Trigger Conditions' and 'Trigger Actions' on the right. The 'Settings' section includes fields for 'Title' (Alert Title), 'Description' (Alert Description), 'Search' (SPL Query), 'App' (Search & Reporting (search)), 'Permissions' (Private), 'Alert type' (Scheduled), 'Time Range' (Last 2 minutes), and 'Cron Expression' (\*/\*/\*/\*/\*). The 'Trigger Conditions' section includes 'Trigger alert when' (Number of Results), 'is greater than' (0), 'Trigger' (Once), and 'Throttle' (unchecked). The 'Trigger Actions' section includes 'When triggered' (Action icon Send email, Add to Triggered Alerts).

- **Alert Title and Alert Description:** Defines the name and a brief summary of the alert's purpose.
- **SPL Query:** The event detection logic used to scan data and generate an alert when specific patterns are found.
- **Alert Type and Time Range:** Sets a cron job for the alert query to run at specific time intervals, checking events within a designated time window.
- **Trigger Condition:** Determines exactly when an alert should be triggered based on specified parameters (e.g., result count is greater than zero).
- **Trigger Action:** The actions performed once an alert is triggered (e.g., sending an automated email notification).

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## Alert Conditions

- Alerts are set to execute every 2 minutes.
- Trigger Condition: Results > 0.
- Email notifications were automatically sent to the mailbox whenever suspicious activity was detected.
- Ten Scheduled Alerts were created which triggers upon matching the SPL queries.

| Searches, Reports, and Alerts                                                                                                                                                                                  |                                          |       |                     |              |       |        |        |         |          |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------|---------------------|--------------|-------|--------|--------|---------|----------|--|
| Searches, reports, and alerts are saved searches created from pivot or the search page. <a href="#">Learn more</a>                                                                                             |                                          |       |                     |              |       |        |        |         |          |  |
| 10 Searches, Reports, and Alerts    Type: All    App: Search & Reporting (search)    Owner: Administrator (admin)    filter    10 per page                                                                     |                                          |       |                     |              |       |        |        |         |          |  |
| Name                                                                                                                                                                                                           | Actions                                  | Type  | Next Scheduled Time | Display View | Owner | App    | Alerts | Sharing | Status   |  |
| <b>Host Discovery Commands Executed</b><br>Detects execution of common host discovery and user enumeration commands that may indicate post-exploitation reconnaissance.                                        | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>Multiple Failed Login Attempts</b><br>Detects 5 or more failed Windows login attempts within one minute, indicating possible brute-force activity.                                                          | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>Network Scan Activity</b><br>Detects hosts attempting connections to multiple destination ports within a short time window, indicating possible network reconnaissance or Nmap scanning.                    | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>New User Account Created</b><br>Detects creation of a new user account, which may indicate unauthorized account creation or persistence.                                                                    | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>Obfuscated PowerShell Command Detected</b><br>Detects PowerShell execution using common obfuscation and defense evasion techniques frequently observed in post-exploitation activity.                       | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>PowerShell Execution Detected</b><br>Detects execution of PowerShell, which may indicate administrative activity or malicious post-exploitation behavior.                                                   | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>Registry Key Modified</b><br>Detects modifications to Windows Registry persistence locations that may indicate malware installation or attacker persistence.                                                | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>Scheduled Task Created</b><br>Detects creation of a Windows scheduled task, which may indicate persistence or unauthorized automated execution.                                                             | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>Successful Login After Multiple Failed Attempts</b><br>Detects a successful Windows login following five or more failed login attempts within minutes, indicating a possible successful brute-force attack. | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |
| <b>User Added to Administrator Group</b><br>Detects when a user account is added to the local Administrators group, which may indicate privilege escalation or persistence.                                    | <a href="#">Edit</a> <a href="#">Run</a> | Alert | none                | none         | admin | search | 0      | App     | Disabled |  |

**Alert Title:** Host Discovery Commands Detected

**Description:** Detects execution of common host discovery and user enumeration commands that may indicate post-exploitation reconnaissance.

**SPL:**    index=sysmon EventCode=1  
          (CommandLine=whoami OR CommandLine=hostname)  
          | table \_time User CommandLine ParentImage

**Time Range:** Last 2 Minutes

**Cron Expression:** \*/2 \* \* \* \* (Executes Every 2 Minutes)

**Trigger Condition:** Number of Results is greater than 0

**Trigger:** Once

**Trigger Action:** Send Email

*Note: To preserve the readability of the report only one alert is explained in detail.*

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## ATTACK SIMULATION

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A realistic attack was conducted from Kali Linux against the Windows 10 endpoint in a controlled lab environment.

### 1. Reconnaissance

The Attacker performed TCP port scan.

```
(kali㉿kali)-[~]
└─$ sudo nmap -sT 192.168.1.12
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-03 23:12 +0530
Nmap scan report for 192.168.1.12
Host is up (0.0016s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp    open  msrpc AGYADWAS
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:8C:0C:80 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.56 seconds
```

### 2. Brute Force

Attacker attempted multiple SSH login attempts using Hydra.

```
(kali㉿kali)-[~]
└─$ hydra -l admin -P password.txt ssh://192.168.1.12
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-03 23:13:18
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 10 tasks per 1 server, overall 10 tasks, 10 login tries (1:1/p:10), ~1 try per task
[DATA] attacking ssh://192.168.1.12:22/
[22][ssh] host: 192.168.1.12 login: admin password: admin123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-03 23:13:20
```

### 3. Initial Access

Attacker successfully authenticates using found credentials.

```
(kali㉿kali)-[~]
└─$ ssh admin@192.168.1.12
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
admin@192.168.1.12's password:

Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

admin@DESKTOP-TLRUEGI C:\Users\admin>
```

### 4. Discovery

Attacker uses common Discovery commands to collect system information.

```
admin@DESKTOP-TLRUEGI C:\Users\admin>whoami
desktop-truegi\admin

admin@DESKTOP-TLRUEGI C:\Users\admin>hostname
DESKTOP-TLRUEGI
```

# SIEM LAB – SPLUNK ENTERPRISE

## 5. PowerShell Execution

Attacker Launches Powrshell via commandline.

```
admin@DESKTOP-TLRUEGI C:\Users\admin>powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\admin>
```

## 6. Encoded PowerShell Command Execution

Attacker Executed Encoded Powershell command.

```
PS C:\Users\admin> powershell -EncodedCommand RwB1AHQALQBDAG8AbQBwAHUAdAB1AHIASQBwAGYAbwA=

WindowsBuildLabEx           : 19041.1. amd64fre.vb release. 191206-1406
WindowsCurrentVersion       : 6.3
WindowsEditionId            : Core
WindowsInstallationType     : Client
WindowsInstallDateFromRegistry : 27-06-2026 10:32:55
WindowsProductId           : 00326-10000-00000-AA538
WindowsProductName         : Windows 10 Home
WindowsRegisteredowner     : Max
WindowsSystemRoot          : C:\Windows
...
```

## 7. Privilege Escalation

A new user account name systemuser has been created and added to Administrators group.

```
admin@DESKTOP-TLRUEGI C:\Users\admin>net user systemuser password123 /add
The command completed successfully.

admin@DESKTOP-TLRUEGI C:\Users\admin>net localgroup Administrators systemuser /add
The command completed successfully.
```

## 8. Persistence

Scheduled task and Registry Key modification has been performed for Persistence.

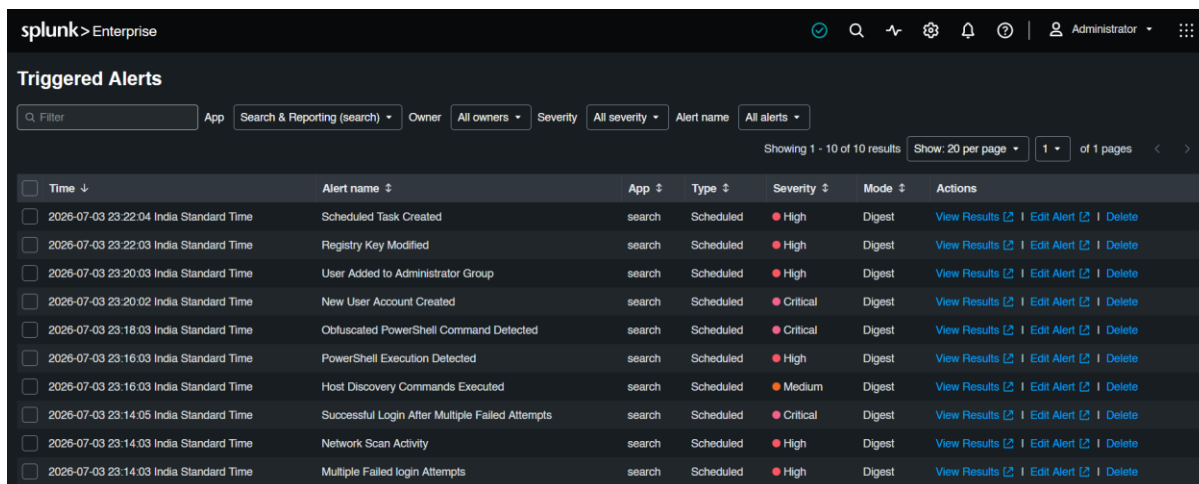
```
admin@DESKTOP-TLRUEGI C:\Users\admin>schtasks /create /tn WindowsUpdateService /tr "C:\Windows\System32\notepad.exe"
/sc ONLOGON /ru admin
SUCCESS: The scheduled task "WindowsUpdateService" has successfully been created.

admin@DESKTOP-TLRUEGI C:\Users\admin>reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v WindowsUpdate
/t REG_SZ /d "C:\Windows\System32\notepad.exe" /f
The operation completed successfully.
```

# SIEM LAB – SPLUNK ENTERPRISE

## INCIDENT INVESTIGATION

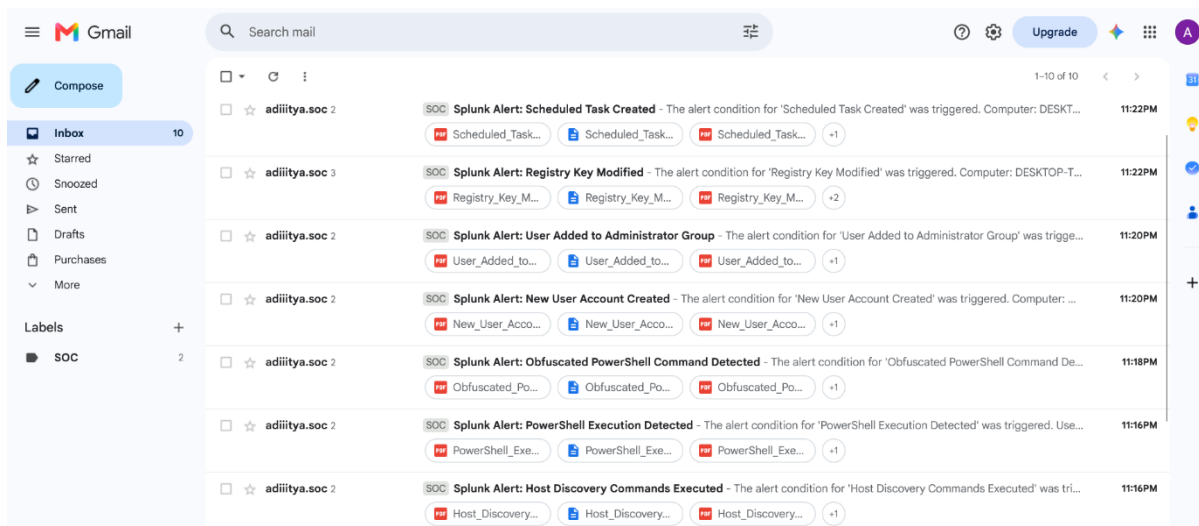
Following the Attack, Alerts were triggered across Splunk and Email.



The screenshot shows the Splunk Enterprise interface with the 'Triggered Alerts' section. It displays a table of alerts with columns for Time, Alert name, App, Type, Severity, Mode, and Actions. The alerts are sorted by time, showing various security events such as 'Scheduled Task Created', 'Registry Key Modified', 'User Added to Administrator Group', 'New User Account Created', 'Obfuscated PowerShell Command Detected', 'PowerShell Execution Detected', 'Host Discovery Commands Executed', 'Successful Login After Multiple Failed Attempts', 'Network Scan Activity', and 'Multiple Failed login Attempts'.

| Time                                    | Alert name                                      | App    | Type      | Severity | Mode   | Actions                                                                        |
|-----------------------------------------|-------------------------------------------------|--------|-----------|----------|--------|--------------------------------------------------------------------------------|
| 2026-07-03 23:22:04 India Standard Time | Scheduled Task Created                          | search | Scheduled | High     | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:22:03 India Standard Time | Registry Key Modified                           | search | Scheduled | High     | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:20:03 India Standard Time | User Added to Administrator Group               | search | Scheduled | High     | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:20:02 India Standard Time | New User Account Created                        | search | Scheduled | Critical | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:18:03 India Standard Time | Obfuscated PowerShell Command Detected          | search | Scheduled | Critical | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:16:03 India Standard Time | PowerShell Execution Detected                   | search | Scheduled | High     | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:16:03 India Standard Time | Host Discovery Commands Executed                | search | Scheduled | Medium   | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:14:05 India Standard Time | Successful Login After Multiple Failed Attempts | search | Scheduled | Critical | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:14:03 India Standard Time | Network Scan Activity                           | search | Scheduled | High     | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |
| 2026-07-03 23:14:03 India Standard Time | Multiple Failed login Attempts                  | search | Scheduled | High     | Digest | <a href="#">View Results</a> <a href="#">Edit Alert</a> <a href="#">Delete</a> |

SPLUNK GUI Triggered Alerts



The screenshot shows a Gmail inbox with several email alerts from Splunk. The alerts are categorized by 'SOC' and include details about the triggered conditions and the computer they originated from. The alerts are: 'Splunk Alert: Scheduled Task Created', 'Splunk Alert: Registry Key Modified', 'Splunk Alert: User Added to Administrator Group', 'Splunk Alert: New User Account Created', 'Splunk Alert: Obfuscated PowerShell Command Detected', 'Splunk Alert: PowerShell Execution Detected', and 'Splunk Alert: Host Discovery Commands Executed'.

| From          | Subject                                                                                                                         | Time    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|---------|
| adiitya.soc 2 | SOC Splunk Alert: Scheduled Task Created - The alert condition for 'Scheduled Task Created' was triggered. Computer: DESK...    | 11:22PM |
| adiitya.soc 3 | SOC Splunk Alert: Registry Key Modified - The alert condition for 'Registry Key Modified' was triggered. Computer: DESKTOP-T... | 11:22PM |
| adiitya.soc 2 | SOC Splunk Alert: User Added to Administrator Group - The alert condition for 'User Added to Administrator Group' was trigge... | 11:20PM |
| adiitya.soc 2 | SOC Splunk Alert: New User Account Created - The alert condition for 'New User Account Created' was triggered. Computer: ...    | 11:20PM |
| adiitya.soc 2 | SOC Splunk Alert: Obfuscated PowerShell Command Detected - The alert condition for 'Obfuscated PowerShell Command De...         | 11:18PM |
| adiitya.soc 2 | SOC Splunk Alert: PowerShell Execution Detected - The alert condition for 'PowerShell Execution Detected' was triggered. Use... | 11:16PM |
| adiitya.soc 2 | SOC Splunk Alert: Host Discovery Commands Executed - The alert condition for 'Host Discovery Commands Executed' was tri...      | 11:16PM |

Email Alerts

Following the alert, the attack was reconstructed using Splunk searches.

The investigation identified:

- Network Scan Activity
- Multiple failed authentication attempts against the admin account.
- Successful SSH authentication.
- Interactive PowerShell execution.
- Encoded PowerShell commands.
- Creation of a privileged local account.
- Addition of the account to the Administrators group.
- Scheduled Task persistence.
- Registry Run Key persistence.

# SIEM LAB – SPLUNK ENTERPRISE

## INCIDENT TIMELINE

The attack timeline was successfully reconstructed using Windows Event Logs and Sysmon telemetry.

| Time  | Activity                | Evidence        | MITRE ATT&CK    |
|-------|-------------------------|-----------------|-----------------|
| 23:12 | Nmap Scan               | Sysmon Event 3  | T1046           |
| 23:13 | Hydra Brute Force       | 4625            | T1110           |
| 23:13 | SSH Login (admin)       | 4624            | T1078           |
| 23:14 | Discovery Commands      | Sysmon Event 1  | T1082 and T1033 |
| 23:15 | PowerShell Execution    | Sysmon Event 1  | T1059.001       |
| 23:16 | Encoded PowerShell      | Sysmon Event 1  | T1059.001       |
| 23:18 | Created systemuser      | 4720            | T1136.001       |
| 23:19 | Added to Administrators | 4732            | T1098           |
| 23:20 | Scheduled Task          | Sysmon Event 1  | T1053.005       |
| 23:20 | Registry Persistence    | Sysmon Event 13 | T1547.001       |

## MITRE ATT&CK MAPPING

All the Adversary Tactics and Techniques are mapped to MITRE ATT&CK using <https://mitre-attack.github.io/attack-navigator/> tool.

| Technique       | Description                                                  |
|-----------------|--------------------------------------------------------------|
| T1046           | Network Service Discovery                                    |
| T1110           | Brute Force                                                  |
| T1078           | Valid Accounts                                               |
| T1082 and T1033 | System Information Discovery and System Owner/User Discovery |
| T1059.001       | PowerShell                                                   |
| T1136.001       | Local Account Creation                                       |
| T1098           | Account Manipulation                                         |
| T1053.005       | Scheduled Task                                               |
| T1547.001       | Registry Run Key                                             |

# SIEM LAB – SPLUNK ENTERPRISE

## INDICATORS OF COMPROMISE (IOC's)

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| IOC Type         | Value                                                            |
|------------------|------------------------------------------------------------------|
| Attacker IP      | 192.168.1.10                                                     |
| Victim IP        | 192.168.1.12                                                     |
| SIEM             | 192.168.1.8                                                      |
| Compromised User | admin                                                            |
| Created User     | systemuser                                                       |
| Scheduled Task   | WindowsUpdateService                                             |
| Registry Key     | HKCU\Software\Microsoft\Windows\CurrentVersion\Run\WindowsUpdate |

## FINDINGS

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**Finding 1:** Multiple failed SSH authentication attempts were detected against the admin account before a successful login.

**Finding 2:** PowerShell was launched shortly after successful authentication, indicating post-compromise activity.

**Finding 3:** A new privileged account was created and added to the local Administrators group.

**Finding 4:** Persistence was established using both Scheduled Tasks and Registry Run Keys.

**Finding 5:** Splunk successfully correlated Windows Event Logs and Sysmon telemetry to reconstruct the complete attack chain.

## RECOMMENDATIONS

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- Implement strong password policies and account lockout thresholds.
- Restrict remote administrative access.
- Enable Multi-Factor Authentication (MFA) where applicable.
- Continuously monitor PowerShell activity with Sysmon.
- Review scheduled tasks and registry autoruns regularly.
- Tune SIEM alerts to reduce false positives.
- Conduct periodic reviews of privileged group memberships.



# SIEM LAB – SPLUNK ENTERPRISE

## LESSONS LEARNED

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- Native Windows logs provide valuable authentication and account management events.
- Sysmon significantly enhances endpoint visibility by capturing process, network, and registry activity.
- Correlating multiple log sources enables accurate attack reconstruction.
- Mapping detections to the MITRE ATT&CK framework improves reporting and detection coverage.
- Automated alerts combined with well-designed dashboards improve SOC response efficiency.

## CONCLUSION

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This project successfully demonstrated the deployment of Splunk Enterprise as a centralized SIEM platform for Windows endpoint monitoring. Custom detection rules, dashboards, and automated email alerts were implemented to identify common attacker behaviours. A realistic attack simulation validated the effectiveness of the detections, and the resulting investigation reconstructed the full attack lifecycle using Windows Event Logs and Sysmon telemetry. The project provided practical experience in SIEM deployment, detection engineering, threat hunting, and incident response, closely reflecting real-world SOC operations.